

Interesting

Reference Designs OF CAMERAS

Title of Reference Design	13MP PDAF UVC Camera	360 Smart Camera
Image		
Key Attractions	The reference design kit for a 13MP phase-detection auto-focus (PDAF) UVC camera supports outputs up to 16 megapixels, offering speed, low latency, and energy efficiency.	The 360 smart camera reference design combines multiple sensors with software and processors, enabling it to support machine learning, capture 360-degree video, and efficiently handle various computing tasks.
Highlights	The THSCU101 is a reference design kit by the collaboration of THine Solutions Inc. and Infineon Technologies featuring a 13MP camera with PDAF capabilities. It integrates Infineon's EZ-USB CX3 USB 3.0 camera controller and THine's THP7312 image signal processor (ISP), forming an advanced imaging solution. At its core is a 13MP camera module with PDAF, using Sony's IMX258 CMOS image sensor. The kit is further equipped with ISP firmware, which standardises image quality across various CMOS camera modules, ensuring consistent high-quality image output. The kit offers plug-and-play functionality and firmware specifically designed for USB camera operations, making the development process more efficient and reducing engineering costs. It serves as both a ready-to-use product and a development tool for creating advanced camera systems, suitable for developers. It offers finely tuned image quality with automatic adjustments for focus, white balance, and exposure to ensure the best possible image output. The system includes a range of connectivity options like an I2C master controller, an I2S master for audio, a UART supporting up to 4 Mbps, and a 33MHz SPI master. It also features twelve GPIOs and is designed for ultra-low power consumption in its core power-down mode.	The Qualcomm 360 smart camera reference design incorporates the Qualcomm Vision Intelligence Platform, a system that offers good image processing and advanced artificial intelligence features within an economical framework. The Vision Intelligence 300 is based on the QCS603 SoC, and the Vision Intelligence 400 on the QCS605 SoC, both accommodating Linux or Android operating systems and integrating power management ICs, audio codec support, and Wi-Fi/Bluetooth connectivity. This integration streamlines the bill of materials and simplifies commercialisation. The platform includes a dual 14-bit Qualcomm Spectra 270 image signal processor (ISP) supporting dual 16MP sensors, an advanced 10nm FinFET process for thermal and power efficiency, the Qualcomm Adreno 615 GPU reaching up to 780MHz, and the Qualcomm Hexagon 685 DSP for running deep neural networks and complex computations. Robust security features encompass secure boot from a hardware root of trust, a trusted execution environment, hardware crypto engines, storage security, debug security with lifecycle control, key provisioning, and wireless protocol security.
Applications	The reference design is suitable for body cameras, scanners, surveillance systems, webcams, AR eyewear, biometric systems, visual aid glasses, microscopes, medical endoscopes, and equipment for machine vision.	The reference design is suitable for action cameras, VR/360 devices, home and enterprise security systems, and wearable cameras.
OEM Brand	THine Solutions Inc. and Infineon Technologies	Qualcomm
URL	https://www.electronicsforu.com/electronics-projects/13mp-pdaf-uvc-camera-reference-design	https://www.electronicsforu.com/electronics-projects/360-smart-camera-reference-design